A WEAK DECOMPOSITION OF THE CIRCULAR CONE IN QUESTION 2 OF SZUSTERMAN

AGENT LANE 11

Status

This packet gives a candidate counterexample to the support-inclusion formulation of Question 2 in Szusterman, *Extremizers in Soprunov and Zvavitch's Bezout inequalities for mixed volumes* [1]. The question asks whether the cone

$$C = \mathrm{Conv}(e_n, B), \qquad B = \pi_{e_n^\perp}(B_2^n) \subset e_n^\perp,$$

is weakly indecomposable. Under the displayed Definition 7 condition $\mathrm{supp}(S_{K+L}) \subset \mathrm{supp}(S_K)$, the answer is negative: C is weakly decomposable in every dimension $n \geq 3$.

Source question

The source paper defines weak decomposability as follows: a convex body K with non-empty interior is weakly decomposable if there is an L, not homothetic to K, such that

$$\mathrm{supp}(S_{K+L}) \subset \mathrm{supp}(S_K).$$

Immediately after recalling this definition and its consequences, the paper asks:

> Is it true that the cone $C = \mathrm{Conv}(e_n, B)$, where $B = \pi_{e_n^\perp}(B_2^n) \subset e_n^\perp$ is the unit ball in $e_n^\perp$, is weakly indecomposable?

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minimizing $b_2(.)$. The next proposition states that a b_2 minimizer must be indecomposable: this is [SZ, Theorem 3.3].

Proposition 3. *Assume K is decomposable, i.e. $K = A + B$, with A and B not homothetic. Then $b_2(K) > 1$.*

It turns out that Theorem 6 and Proposition 3 are merely special cases of a larger result. In [SSZ2], the concept of *weak decomposability* was introduced.

Definition 7. Let K be a convex body with non-empty interior. We say that K is weakly decomposable, if there exists L, non-homothetic to K, such that $supp(S_{K+L}) \subset supp(S_K)$.

Remark 1 Any smooth convex body is weakly decomposable, because $supp(S_K) = \mathbb{S}^{n-1}$ (take any polytope for L). Likewise, any convex body such that $supp(S_K)$ contains an open subset of $\mathbb{S}^{n-1}$, is weakly decomposable.

Remark 2 The simplex is the only n-polytope which is not weakly decomposable (see Lemma 6 in the Appendix for a proof). This is why Theorem 8 below implies Theorem 6 above.

Remark 3. If K is decomposable, say $K = A + B$ with A, B not homothetic (and therefore, A, K not homothetic), then K is weakly decomposable. Indeed, choose $L = A$. Then:

$$S_{K+L} = S_{2A+B} = \sum_{k=0}^{n-1} \binom{n-1}{k} 2^{n-1-k} \sigma_k \quad \text{while} \quad S_K = \sum_{k=0}^{n-1} \binom{n-1}{k} \sigma_k$$

where $\sigma_k = S(A[n-1-k], B[k], .)$ are the mixed surface area measures between A, B. It follows that $supp(S_K) = supp(S_{K+L}) = \cup_{k=0}^{n-1} supp(\sigma_k)$.

The following theorem is due to Saroglou, Soprunov, and Zvavitch (see [SSZ2, Theorem 5.7]).

Theorem 8. *Let K be a weakly decomposable body. Then $b_2(K) > 1$.*

In other words, like decomposability, weak decomposability is an excluding condition (when searching minimizers of $b_2(.)$). The proof reproduces the induction of Theorem 6 replacing $P_{j,s}$ with Wulff shape perturbations. While the discrete nature of surface area measures of polytopes (and fact 2, i.e. nice behaviour of the support of surface area measures, with respect to Minkowski sum with a perturbation) was a key ingredient for iteratively deducing $\sigma_r = \lambda_t^r \sigma_0$, in the above proof of Theorem 6, in this other setting, fact 2 wouldn't hold (if we also choose to replace $P_{i,t}$ with a Wulff-shape perturbation), but comes for free via the assumption $S_{K+L} << S_K$, and the main ingredient to derive $\sigma_r = \lambda_t^r \sigma_0$ (where now $\sigma_r = S(L[r], K[n-1-r], .)$), is then Alexandrov's variational lemma (see Lemmas 1 and 2). For more details, see [SSZ2], p.16.

In view of the above theorem, it was asked in [SSZ2] whether n-simplices are the only convex bodies $K \in \mathcal{K}_0^n$, which are not weakly decomposable. If true, this would solve Conjecture 1.

Question 2. *Is it true that the cone $C = Conv(e_n, B)$, where $B = \pi_{e_n^\perp}(B_2^n) \subset e_n^\perp$ is the unit ball in $e_n^\perp$, is weakly indecomposable ?*

4. An excluding condition with isoperimetric ratios

Definitions and notation

Write points of $\mathbb{R}^n$ as (x,t), with $x \in H := e_n^\perp \simeq \mathbb{R}^{n-1}$ and $t \in \mathbb{R}$. Then

$$C = \{(x,t) : 0 \le t \le 1,\ \|x\|_2 \le 1-t\}.$$

Its support function is

$$h_C(\xi, s) = \max\{\|\xi\|_2, s\}, \qquad (\xi, s) \in H \times \mathbb{R}.$$

The surface area measure of C is supported on

$$\Omega_C = \{-e_n\} \cup \left\{ \frac{(\theta, 1)}{\sqrt{2}} : \theta \in \mathbb{S}^{n-2} \subset H \right\}.$$

The point $-e_n$ is the base normal; the spherical copy of $\mathbb{S}^{n-2}$ is the set of normals to the lateral conical surface. The apex normal cone and the normal cones of the base rim carry no $(n-1)$-dimensional boundary area.

Proof intuition

The circular cone has a large zero-area normal cone at its apex. Sweeping the whole cone by a small horizontal segment turns the apex into a horizontal segment, but that top segment still has codimension at least one inside the boundary and hence contributes no surface area measure. At height t, the new body has horizontal section equal to a fixed segment plus a Euclidean ball of radius $a-t$. Thus all genuine side normals remain exactly the old lateral cone normals $(\theta, 1)/\sqrt{2}$, and the only facet normal remains the base normal $-e_n$. The sweep is therefore a non-homothetic Minkowski summand that does not enlarge the support of the surface area measure.

Counterexample

Theorem 1. *For every $n \geq 3$, the cone*

$$C = \mathrm{Conv}(e_n, \pi_{e_n^\perp}(B_2^n))$$

is weakly decomposable in the sense of Definition 7 of [1]*. Consequently Question 2 of* [1] *has a negative answer.*

Proof. Let $H = e_n^\perp$ and choose a nondegenerate compact segment

$$P = [-v, v] \subset H, \qquad v \neq 0.$$

Fix $\varepsilon > 0$, and put

$$L = P + \varepsilon C.$$

Then L is a full-dimensional convex body because εC is full-dimensional. Also L is not homothetic to C: in direction e_n, the exposed face C^{e_n} is the single apex $\{e_n\}$, whereas

$$L^{e_n} = P + \varepsilon e_n$$

is a nondegenerate segment.

It remains to show that $C + L$ has no surface area measure outside $\mathrm{supp}(S_C)$. Since

$$C + L = P + (1+\varepsilon)C,$$

it is enough to analyze bodies of the form $K_a := P + aC$, $a > 0$. Their support functions are

$$h_{K_a}(\xi, s) = h_P(\xi) + a\max\{\|\xi\|_2, s\}, \qquad (\xi, s) \in H \times \mathbb{R}. \tag{1}$$

Equivalently,

$$K_a = \{(x,t) : 0 \leq t \leq a,\ x \in P + (a-t)B_H\}, \tag{2}$$

where B_H is the Euclidean unit ball in H. Formula (2) makes the boundary geometry explicit.

The base $t = 0$ is the only $(n-1)$-dimensional facet with horizontal normal, and its outer normal is $-e_n$. The top set $P + ae_n$ has dimension $1 \leq n-2$, so its normal cone contributes zero to S_{K_a}. Similarly, the lower rim of the base has dimension at most $n-2$, so its normal cone also contributes zero surface area.

On the remaining boundary, $0 < t < a$ and x lies on the boundary of $P + (a-t)B_H$. At every regular such point the horizontal outer normal is some $\theta \in \mathbb{S}^{n-2}$, and locally the supporting equation is

$$\langle x, \theta\rangle + t = h_P(\theta) + a.$$

Thus the corresponding outer unit normal in $\mathbb{R}^n$ is

$$\frac{(\theta,1)}{\sqrt{2}},$$

one of the original lateral normals of C. At nonregular side points, possibly more than one horizontal normal θ is active, but every supporting normal that occurs is still of the displayed lateral form. Any positive-area flat side pieces therefore have normals in the same lateral sphere, and the lower-dimensional intersections between such pieces carry no $(n-1)$-dimensional boundary area.

Therefore

$$\operatorname{supp}(S_{K_a}) \subset \{-e_n\} \cup \left\{ \frac{(\theta,1)}{\sqrt{2}} : \theta \in \mathbb{S}^{n-2} \right\} = \operatorname{supp}(S_C).$$

Taking $a = 1+\varepsilon$ gives

$$\operatorname{supp}(S_{C+L}) = \operatorname{supp}(S_{P+(1+\varepsilon)C}) \subset \operatorname{supp}(S_C),$$

with L non-homothetic to C. This is exactly weak decomposability of C, so the proposed weak indecomposability fails. □

Verification notes

- Local lightweight indexes were searched for `2304.00366`, `Soprunov`, `Zvavitch`, `weakly decomposable`, `weakly indecomposable`, `b_2(K)`, and the circular-cone keywords. No duplicate packet or prior attempt for this exact cone question was found.
- Bounded web search on 2026-07-18 used the exact phrases `weakly indecomposable Soprunov Zvavitch cone`, `weakly decomposable Conv(e_n`, `Soprunov-Zvavitch weakly decomposable cone`, and the source title. It found the source article and later general Bezout-type inequality papers, but no separate later paper explicitly resolving this cone question.
- No computational verification is needed. The proof is the explicit Minkowski summand $L = P + \varepsilon C$ and the face/support calculation above.

Limitations

This packet answers only the support-inclusion version of Question 2 about the displayed circular cone. The same source discussion also refers to the stronger condition $S_{K+L} \ll S_K$ from Saroglou–Soprunov–Zvavitch; the present segment-sweep construction is not claimed to settle that absolute-continuity variant. It also does not settle the broader conjecture that $b_2(K) = 1$ characterizes simplices for all convex bodies in dimensions $n \geq 4$, and it does not classify all weakly indecomposable convex bodies.

Human review recommendation

Review as a likely-valid counterexample. The main points to check are: whether the intended target is the displayed support-inclusion definition rather than the stronger absolute-continuity variant, the claim that $L = P+\varepsilon C$ is not homothetic to C, and the assertion that all positive-area boundary pieces of $P + aC$ have normals in the old base/lateral support of C.

References

[1] M. Szusterman, *Extremizers in Soprunov and Zvavitch's Bezout inequalities for mixed volumes*, arXiv:2304.00366, 2023. `https://arxiv.org/abs/2304.00366`